



PLANET 6

GJ 1214 B

Description: Warm mini-Neptune wrapped in a global haze so thick that telescopes cannot see down through it, which is why its spectrum looks famously featureless.

Host star: GJ 1214, a small, cool red dwarf less than a fifth of the mass of the Sun.

System: The only planet known around GJ 1214 so far.

Location: Constellation Ophiuchus, about 48 light-years from Earth. Far too faint for the naked eye (magnitude 15).

Discovery: In 2009 by the MEarth Project, a network of small robotic telescopes, led by David Charbonneau et al.

Datasheet

EXOPLANET DATA		
	TEMPERATURE	~280 °C
	PRESSURE	No surface
	ATMOSPHERE	Haze, H ₂ O
	GRAVITY	1.13 g
	PLANET RADIUS	
	ORBITAL PERIOD	
	SEMI-MAJOR AXIS	
	TRANSIT DEPTH	
	TRANSIT DURATION	

HOST STAR DATA		
	NAME	GJ 1214
	SPECTRAL TYPE	M4V
	TEMPERATURE	3101 K
	DISTANCE	48 ly
	MASS	0.182 M _☉
	STAR RADIUS	0.216 R _☉
	AGE	3.0 Gyr
	APPARENT MAGNITUDE	15.10
	NUMBER OF PLANETS	1

SURVIVAL CHALLENGE

Crushing atmospheric pressure

NOTES



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NOTES

PHYSICS

 PLANET RADIUS	
 AI-ASSISTANCE	
 REFLECTIONS	

MATHEMATICS

 ORBITAL PERIOD		 SEMI-MAJOR AXIS	
 AI-ASSISTANCE			
 REFLECTIONS			

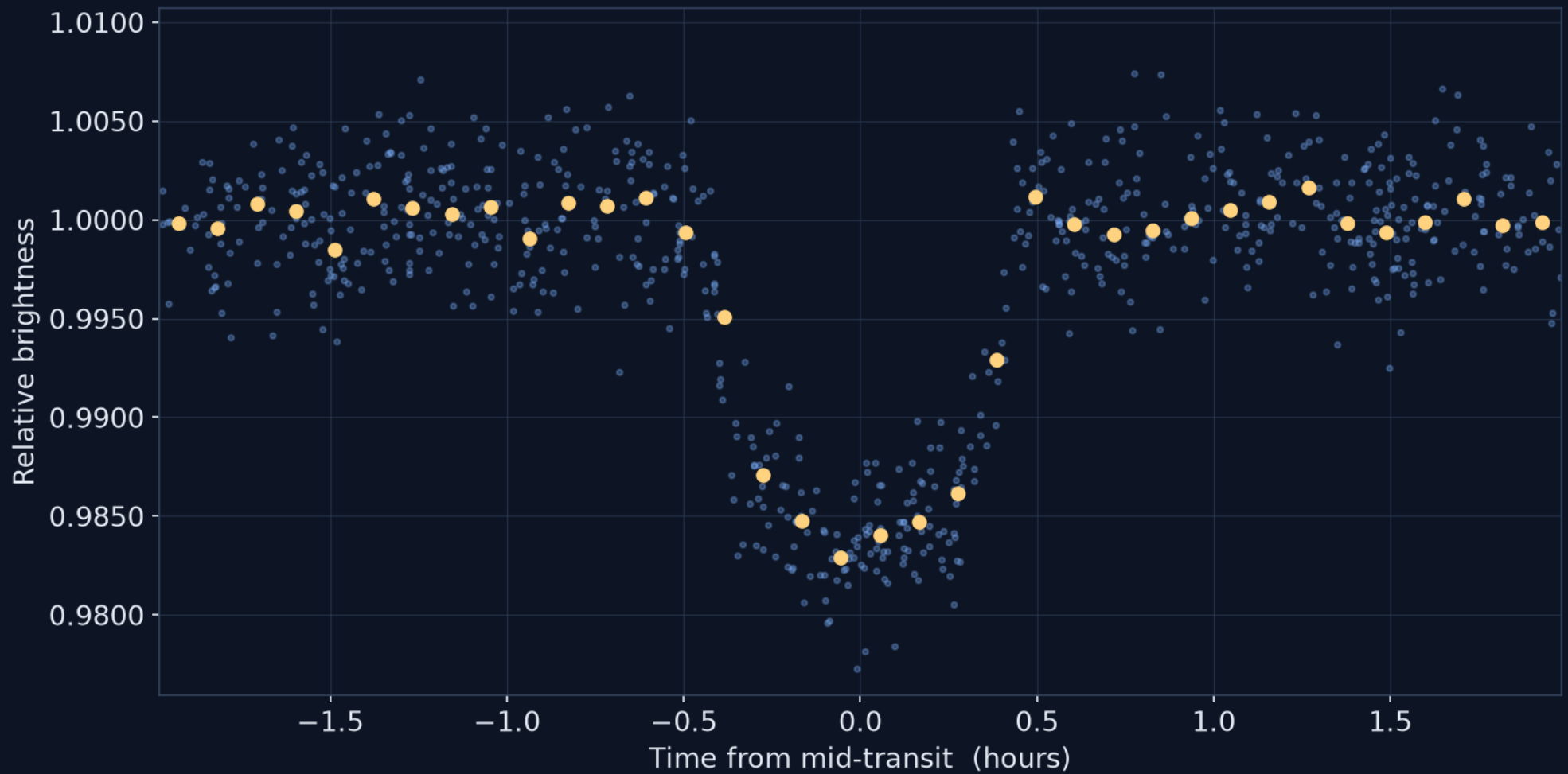
CHEMISTRY

 SAMPLE DATA	
 AI-ASSISTANCE	
 REFLECTIONS	

BIOLOGY

 ORGANISM	
 AI-ASSISTANCE	
 REFLECTIONS	

GJ 1214 b: single transit



Blue points are simulated brightness measurements (many orbits phase-folded together, like real data); the gold points are binned averages.

GJ 1214 b: light curve over several orbits

